
Search for axionlike particle as dark matter candidate with nuclear magnetic resonance

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Abstract

Acknowledgments

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Interesting questions:

- What is the weakest pseudomagnetic field we can measure? What sensitivity in the exclusion plot does it correspond to?

What is nuclear magnetic resonance? What are axions and axionlike particles?

The starting point of this thesis is the observation that NMR is a sensitive probe of magnetic fields, while axionlike particles can be treated as an effective oscillating pseudomagnetic field acting on nuclear spins. This makes NMR a natural readout channel for dark-matter searches in the low-field regime.

Astrophysical evidence for dark matter comes from observations such as galaxy rotation curves: the visible matter alone does not explain the measured dynamics. Axions and axionlike particles are attractive candidates because they are weakly coupled, can occupy a large parameter space, and may generate detectable spin-precession signatures through non-gravitational interactions.

The purpose of this introductory chapter is therefore to connect the language of NMR with the language of axion searches. The later chapters use that connection to motivate the detector design, the scan strategy, and the analysis of the measured signal.

1.1 Nuclear magnetic resonance

Nuclear magnetic resonance is a technique for studying nuclear spins in magnetic fields. In the presence of a leading field, spin states split by the Zeeman effect and a polarized sample acquires a net magnetization that precesses at the Larmor frequency. Because the resonant frequency is directly proportional to the magnetic field, NMR is a precision tool for field measurement.

For the purposes of this thesis, the most important features of NMR are its sensitivity to weak fields, its applicability from zero to very high field, and its rich time-domain response. Those properties make it possible to reinterpret axionlike spin precession as a small perturbation on an otherwise standard NMR experiment.

1.1.1 Zeeman splitting and polarization

In a magnetic field, a spin-1/2 system splits into two energy levels. At thermal equilibrium, the populations are slightly imbalanced, producing a polarization proportional to the magnetic field and inversely proportional to temperature. This equilibrium polarization is the origin of the net magnetization used in NMR.

The practical consequence is that stronger leading fields generally improve polarization, but even at high field the thermal imbalance remains small. Hyperpolarization methods can therefore be extremely valuable when signal strength is limited by the available spin population rather than by detector noise.

1.1.2 Spin-projection noise

Even in the absence of a leading field, an ensemble of many spins does not cancel perfectly. Statistical fluctuations leave a residual moment of order $\sqrt{N}\mu$, which is known as spin-projection noise. This sets a fundamental scale for very low-noise measurements.

SPN is usually negligible in ordinary NMR, but it becomes relevant when the signal amplitude is reduced to the point where statistical fluctuations and instrumental noise are comparable. In the axion-search context, it is one of the natural reference scales for the smallest measurable spin signal.

1.1.3 Relaxations

Relaxation describes the return of the spin ensemble toward equilibrium after it has been perturbed. Longitudinal relaxation T_1 controls how quickly the magnetization along the leading field rebuilds, while transverse relaxation T_2^* determines how quickly coherent precession dephases.

These time scales are central to the rest of the thesis because they set the window over which the signal can be observed coherently. In particular, the weakly driven axion signal is only useful if its coherence time and the detector linewidth are compatible with the relaxation times of the sample.

T_1 relaxation

Longitudinal relaxation governs how the population difference between spin states rebuilds after perturbation. In practice, a short T_1 is helpful for rapid repetition, while a long T_1 can preserve polarization for longer measurements.

T_2^* relaxation

Transverse relaxation combines microscopic spin-spin interactions and macroscopic field inhomogeneity. It determines the decay of the free-induction signal and directly controls the linewidth in the frequency domain. For field searches, a narrower linewidth can improve sensitivity, but only if the coherence time remains long enough to integrate signal power efficiently.

Cross relaxation

Cross relaxation is one of the mechanisms by which energy can be exchanged between coupled spin species or spin reservoirs. It is relevant whenever the sample or environment contains more than one spin system, because it can alter the effective relaxation rates seen in the measurement.

1.1.4 Chemical shift

Chemical shielding changes the local magnetic field felt by a nucleus and therefore shifts the resonance frequency. The shift is typically small on an absolute scale, but it can be crucial in spectroscopy because it makes nuclei in different chemical environments distinguishable.

In the current thesis, chemical shift matters mainly as a source of spectral structure and as a reminder that the resonant response is not determined by the external field alone.

1.1.5 Spin-spin coupling

Coupling between nuclear spins splits resonance lines into multiplets. This effect is useful rather than harmful when the goal is signal verification, because the pattern of peaks can identify a particular molecular species and help distinguish genuine NMR signals from noise spikes.

For the calibration sample used later in the thesis, J-coupling provides a clear spectral signature that is much easier to recognize than a single unresolved line.

1.1.6 Experimental schemes

Two pulse-based schemes are especially useful. Free induction decay measures the coherent response after a tipping pulse, while spin echo refocuses dephasing and gives access to the intrinsic transverse relaxation time. Both are diagnostic tools that help characterize the apparatus before an axion measurement.

Free decay

After a $\pi/2$ pulse, the magnetization is tipped into the transverse plane and then decays with the dephasing time T_2^* . The resulting free-induction signal is the simplest way to measure resonance frequency and linewidth.

Spin echo

Spin echo uses a π pulse to reverse dephasing from static field inhomogeneity. By recovering the signal at the echo time, it separates true spin-spin relaxation from broadening caused by imperfect field homogeneity.

1.2 Dark matter hypothesis

The concept of dark matter arose from astronomical observations that could not be explained by visible matter and standard gravity alone. Galaxy rotation curves are the classic example: the outer parts of galaxies rotate too quickly for the observed baryonic mass to provide the required gravitational binding.

This empirical gap motivated a broad search for non-luminous matter. For this thesis, the important point is not the historical detail but the fact that dark matter must interact with ordinary matter in some way if it is to be detected directly.

1.2.1 History

From early galaxy-dynamics studies to later lensing and large-scale structure measurements, the dark-matter hypothesis was strengthened by evidence from many independent observations. The basic conclusion is that a large fraction of the matter content of the universe is not directly visible.

1.2.2 Recent developments

Modern dark-matter searches span weakly interacting particles, ultralight bosons, and direct-detection strategies based on precision measurement. Axions and axionlike particles are especially interesting because they connect cosmology with laboratory-scale experiments through tiny but coherent effects.

1.3 Axions and axionlike particles

The QCD axion was originally proposed as a solution to the strong CP problem. Axionlike particles generalize the idea to a broader class of light bosons whose masses and couplings are not fixed by the QCD mechanism alone.

The relevance of axions to this thesis is that they can generate oscillating effective fields with frequencies set by the particle mass. That makes them accessible to resonant experiments such as NMR.

1.3.1 QCD axion

The QCD axion is a hypothetical pseudoscalar particle that appears when a global symmetry is introduced to explain why strong interactions conserve CP so well. Its properties are tightly linked to the QCD scale, which makes it a special case rather than the most general one.

1.3.2 Axionlike particles

Axionlike particles share the oscillatory and weakly coupled character of the QCD axion, but their mass range can be much wider. This flexibility makes them attractive targets for laboratory searches.

1.3.3 Non-gravitational interactions

The most relevant non-gravitational couplings for this thesis are the ones to photons, gluons, and fermions. Axion-fermion couplings are particularly interesting for spin-precession experiments because they can appear as an effective interaction with nuclear spins.

including *Experiment design for axion research*

1.4 CASPEr project

CASPEr is a family of proposed experiments that use nuclear spins to search for axion-induced effects. The idea is to translate the tiny axion coupling into a measurable precession or phase shift in a carefully controlled NMR setup.

The project has several branches, including electric and gradient-based searches, and both low-field and high-field implementations. The low-field regime is especially relevant when the goal is to scan a wide frequency band with high sensitivity to weak couplings.

1.4.1 Idea

The central idea is to tune the NMR resonance to a frequency that matches the axion Compton frequency. If the two frequencies align, the axion field can act like a tiny driving field and induce a coherent response in the sample.

In practice, the method reduces to a frequency scan: either the signal appears at a particular resonance or the absence of a signal constrains the coupling strength over the scanned band.

CASPEr-E

CASPEr comagnetometers

CASPEr LF, HF

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Axionlike-particle-induced spin precession

Key questions to be answered:

- How is the interaction between axionlike particles and spins?
- What does the axionlike-particle-induced spin precession signal look like?
- What is the optimal way to detect the signal?

How is the interaction between axionlike particles and spins? ...questions here. By answering these questions in this chapter, we can characterize the signal and determine a good strategy for the experiment.

Part of the content related to the frequency scanning is adapted from the Ref. [1]

2.1 Stochasticity

2.2 Periodic modulations

2.2.1 Daily modulation

2.2.2 Annual modulation

2.3 Spin-precession signal

2.3.1 Relaxations and coherence

2.3.2 Estimation

2.3.3 Numerical simulation

Low frequency: long ALP coherence time, short measurement time

medium frequency

high frequency: short ALP coherence time, short $T_{2\text{star}}$, long T_2 , long T_1

considering the T_1 relaxation of hyperpolarized Xenon, we have ... factor of 1/3 worse sensitivity.

2.3.4 Summary

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Sample and Apparatus

Interesting questions to be answered:

- What are the experimental apparatus?
- What are the uses of them?
- What happens during the experiment?

meta

3.1 Overview

The apparatus can be organized into three functional blocks: signal generation, signal reception, and initial processing. The first block prepares the spin system, the second converts magnetic response into an electrical signal, and the third filters and digitizes that signal for later analysis.

This structure mirrors the actual workflow of the experiment. The magnet and excitation hardware define the operating point, the pickup and SQUID chain set the sensitivity, and the processing electronics decide how much of the raw signal can be recovered from noise.

3.2 Magnets

3.3 Coil system

kitchen roll

3.4 Detection chain

mention 1- and 3-coil pickups.

3.5 Data processing

4

Results and analysis

Key questions to be answered:

-
-
-

5

Conclusions & Outlook

Key questions to be answered:

-
-
-

Key questions to be answered:

-
-
-

???

A.1 Hypothesis test

Key questions to be answered:

- How to set up hypothesis?
-
-

A.2 The Neyman–Pearson Lemma

The *Neyman–Pearson lemma* is a foundational result in statistical hypothesis testing. It establishes the form of the most powerful test for discriminating between two simple hypotheses at a fixed significance level.

A.2.1 Problem Setup

Suppose we observe data x drawn from a probability distribution. We wish to test

$$H_0 : x \sim f_0(x)$$

against

$$H_1 : x \sim f_1(x),$$

where both $f_0(x)$ and $f_1(x)$ are fully specified probability density functions. Such hypotheses are referred to as *simple hypotheses*.

The goal is to construct a statistical test that

- maintains a fixed probability of false alarm (Type I error),
- while maximizing the probability of correctly detecting the alternative hypothesis (power).

A.2.2 Likelihood Ratio

The key quantity is the likelihood ratio

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)}.$$

This ratio quantifies how much more likely the observed data is under the alternative hypothesis than under the null hypothesis.

Large values of $\Lambda(x)$ indicate that the observation is more consistent with H_1 than with H_0 .

A.2.3 Statement of the Neyman–Pearson Lemma

For a fixed significance level α , the most powerful test of H_0 against H_1 is given by

$$\Lambda(x) > k,$$

where the threshold k is chosen such that

$$P(\Lambda(x) > k \mid H_0) = \alpha.$$

Equivalently, the optimal decision rule is:

$$\text{Reject } H_0 \quad \text{if} \quad \frac{f_1(x)}{f_0(x)} > k.$$

The lemma therefore establishes that the likelihood-ratio test is optimal among all tests with the same false-positive rate.

A.2.4 Type I Error and Power

The significance level is defined as

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}),$$

which represents the probability of falsely rejecting the null hypothesis.

The power of the test is

$$1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ true}),$$

which measures the probability of correctly detecting the alternative hypothesis.

The Neyman–Pearson lemma guarantees that the likelihood-ratio test maximizes the power for a fixed α .

A.2.5 Gaussian Example

Consider the case

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

with known variance σ^2 . We test

$$H_0 : \mu = 0$$

against

$$H_1 : \mu = 1.$$

Under the null hypothesis,

$$f_0(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right),$$

while under the alternative hypothesis,

$$f_1(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-1)^2}{2\sigma^2}\right).$$

The likelihood ratio is therefore

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)} = \exp\left[-\frac{(x-1)^2}{2\sigma^2} + \frac{x^2}{2\sigma^2}\right].$$

Expanding the exponent,

$$(x - 1)^2 = x^2 - 2x + 1,$$

so that

$$\log \Lambda(x) = \frac{x}{\sigma^2} - \frac{1}{2\sigma^2}.$$

Hence, rejecting for large $\Lambda(x)$ is equivalent to rejecting for sufficiently large x .

Bibliography

- [1] Yuzhe Zhang, Deniz Aybas Tumturk, Hendrik Bekker, Dmitry Budker, Derek F. Jackson Kimball, Alexander O. Sushkov, and Arne Wickenbrock. **Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments**. *Annalen der Physik* 536:1 (2024), 2300223. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/andp.202300223> (see page 7).